1. Identity

Document	STAR-PKG-STAR-ROOTFIND-EN · v2.1 · 2026-11-01
Package	star-rootfind
Version	0.1.0
Capability	star.numerics.bracketed-root
Maturity	RELEASED
Python	⩾3.9
Source commit	94fcf7c
Source	https://github.com/Andrea07072000/S.T.A.R.-/tree/main/packages/star_rootfind

2. Licence and obligations

Apache License 2.0. You may use, copy, modify and distribute it, also commercially, and you receive a patent licence from the contributors. When you redistribute it: keep the LICENSE file and the copyright notices (and a NOTICE file, if a future version adds one), and state in modified files that you changed them. You may not use the S.T.A.R. name to endorse your product.

3. What it needs at run time

Nothing beyond the Python standard library: it works offline.

4. Integrity

SHA-256 fingerprints of the files published on this website. Check a downloaded file with:

```
python -c "import hashlib,sys;print(hashlib.sha256(open(sys.argv[1],'rb').read()).hexdigest())" star_rootfind-0.1.0-py3-none-any.whl
```

FILE	SHA-256
star_rootfind-0.1.0-py3-none-any.whl	87a20d9ca095d31b6d29a3daa05ddb5454c20fa2abbad37c6966f04cbbdf6669
star_rootfind.py	14bd52ede61ac2bc042e09e05ad72f76716eb232a3f5a276958074206fca360b
star_rootfind-0.1.0-source.zip	e49c88a600a81b9da0b9beca51b4a8af3106a638385b6f60c2c9983a5101005f

5. What it does, and its limits

Root of a real function inside a bracket by bisection over the floats (no tolerance, at most 64 evaluations after the two ends, the result within one float of the change of sign) and a scan for every change of sign over equal pieces; 600 problems in six families, Kepler's equation included, agree with mpmath at 50 digits, SciPy brentq and fluids brenth within 2.1e-15 relative.

A change of sign is found, not proved to be a root: for a discontinuous function the result is the jump. The result is exact for f as computed; its distance from the true root is the rounding error of f divided by its slope, which this module cannot know (x*x − 2 on [0, 2] returns the float below sqrt(2) rounded). Roots without a change of sign (double roots) are not found, and `find_roots` misses two roots that fall in the same piece: it is a scan, not a guarantee of completeness. Bisection uses up to 64 evaluations where Brent's method often needs ten: this is for certainty of termination and of the last digit, not for speed. One dimension only.

- `find_root(f, a, b)`, a < b, with f(a) and f(b) of opposite sign (or one of them zero), returns a float of [a, b] such that f changes sign between it and a neighbouring float, or f is exactly zero there.
- The interval is halved in the number of floats it contains: f is evaluated at a, at b, and then at most 64 times, whatever the width of the bracket. `find_roots(f, a, b, pieces)` splits [a, b] into equal pieces and returns, sorted, one root for each piece with a change of sign and every piece edge where f is zero.
- Measured on 600 bracketed problems in six families (cube root, exponential, cosine, Kepler's equation up to e = 0.95, tanh, logarithm): the result agrees with mpmath at 50 digits, SciPy `brentq` and fluids `brenth` within 2.1e-15 relative.
- Every function returns floats or raises `ValueError`: f not callable, limits that are not finite numbers within 1e300 or not in increasing order, no change of sign, a value of f that is not a finite real number, an invalid number of pieces. An exception raised by f itself is not caught.

6. Export, use and certification

The package is published and freely available. That usually takes software out of export-control lists, but not every use of it: read the page on space activities and export control before using it in a controlled programme. It is not certified or qualified for flight; the evidence describes what was tested and how.

7. Citation and bill of materials

```
@software{cavazzini_star_rootfind_0_1_0,
  author  = {Cavazzini, Andrea},
  title   = {{star-rootfind}: Bracketed roots to the last float},
  version = {0.1.0},
  year    = {2026},
  url     = {https://andrea07072000.github.io/S.T.A.R.-/software/star_rootfind/},
  note    = {Apache-2.0. Source: https://github.com/Andrea07072000/S.T.A.R.-, commit 94fcf7c}
}
```

CITATION.cff · BibTeX · SBOM CycloneDX 1.5 · SBOM SPDX 2.3

8. Warranty and liability

Provided "as is", without warranties or conditions of any kind (Apache License 2.0, sections 7 and 8), to the extent the law allows. This never excludes liability that the law does not allow to exclude.

Related: space activities and export control · acceptable use · evidence and limits · licence texts